

Platelet-type Bleeding Disorder 16 (BDPLT16): Comprehensive Disease Characteristics Report

Disease: Platelet-type Bleeding Disorder 16 (BDPLT16) **Also known as:** ITGA2B/ITGB3-related macrothrombocytopenia; Glanzmann thrombasthenia-like syndrome (GTLS); autosomal dominant macrothrombocytopenia with α IIB β 3 gain-of-function **OMIM:** #187800 · **MONDO:** MONDO:0008552 · **ICD-10:** D69.1 · **Category:** Mendelian (autosomal dominant)

Summary

Platelet-type Bleeding Disorder 16 (BDPLT16) is a rare, autosomal dominant inherited platelet disorder defined at the molecular level by heterozygous **gain-of-function** variants in *ITGA2B* (encoding integrin α IIB/GPIIb) or *ITGB3* (encoding integrin β 3/GPIIIa). These variants cluster in the **membrane-proximal (transmembrane and cytoplasmic) region** of the α IIB β 3 integrin and disrupt the conserved intracytoplasmic **salt bridge between α IIB-Arg995 and β 3-Asp723**. Loss of this clasp releases the integrin from its resting conformation, producing a **constitutively active** α IIB β 3 receptor. This is the mechanistic inverse of classic Glanzmann thrombasthenia (GT), which is caused by recessive loss-of-function of the same genes.

Constitutive α IIB β 3 activation drives permanent "outside-in" signaling in megakaryocytes, which perturbs cytoskeletal (actin/tubulin) dynamics and produces **abnormal proplatelet formation**. The clinical consequence is a **macrothrombocytopenia** — large platelets present in reduced numbers — together with reduced α IIB β 3 surface expression (from receptor internalization), a **partial (not absent) platelet aggregation defect**, and lifelong, usually **mild-to-moderate mucocutaneous bleeding** (easy bruising, epistaxis, gum bleeding, menorrhagia). Enlarged and fused α -granules are seen ultrastructurally. Because the count is low and the platelets are large, patients are **frequently misdiagnosed as immune thrombocytopenia (ITP)** and treated ineffectively with corticosteroids/splenectomy.

BDPLT16 is genetically and mechanistically distinct from recessive GT; diagnosis rests on recognizing dominant inheritance, macrothrombocytopenia with *reduced* (not absent) α Ib β 3, a partial aggregation defect, and an activating membrane-proximal *ITGA2B/ITGB3* variant on next-generation sequencing. No disease-specific or curative therapy exists; management is supportive and extrapolated from inherited platelet function disorders — antifibrinolytics (tranexamic acid), recombinant activated factor VII (rFVIIa), platelet transfusion for major bleeds/surgery, and hormonal control of heavy menstrual bleeding. This report synthesizes the molecular pathogenesis, phenotype, genetics, diagnostics, prognosis, treatment, and model systems across all requested disease-characteristic domains.

1. Disease Information

Overview. BDPLT16 is a Mendelian inherited platelet disorder in which a dominant gain-of-function lesion in the fibrinogen receptor α Ib β 3 produces large, poorly functional platelets in reduced numbers, causing a lifelong mild-to-moderate bleeding tendency. It belongs to the broader family of **congenital macrothrombocytopenias** and is a recognized subset now termed **ITGA2B/ITGB3-related macrothrombocytopenia** ([PMID: 41503871](#)).

Key identifiers.

Resource	Identifier
OMIM	#187800 (Bleeding disorder, platelet-type, 16; BDPLT16)
MONDO	MONDO:0008552
ICD-10	D69.1 (Qualitative platelet defects)
Causal genes	<i>ITGA2B</i> (HGNC:6138; OMIM 607759; NCBI Gene 3674; Ensembl ENSG00000005961; UniProt P08514; 17q21.31) · <i>ITGB3</i> (HGNC:6156; OMIM 173470; NCBI Gene 3690; UniProt P05106; 17q21.32)

Synonyms / alternative names. ITGA2B/ITGB3-related macrothrombocytopenia; Glanzmann thrombasthenia-like syndrome (GTLS); autosomal dominant macrothrombocytopenia with α Ib β 3 gain-of-function; α Ib β 3-related macrothrombocytopenia.

Source of information. Information is derived from **aggregated disease-level resources** (OMIM, MONDO) and from **primary clinical/genetic case series and pedigrees** of individual patients — not from population EHR datasets. The evidence base is a set of small multi-generational families reported worldwide (see Section 9).

2. Etiology

Primary cause — genetic (monogenic, dominant, gain-of-function). BDPLT16 is caused by heterozygous gain-of-function variants in *ITGA2B* or *ITGB3*. These are point mutations (and small in-frame deletions) concentrated in the membrane-proximal region of the integrin that destabilize the αIIb -R995/ $\beta 3$ -D723 intracytoplasmic salt bridge, causing constitutive activation (PMID: 29090484; PMID: 29380037; PMID: 33276370).

Genetic risk factors. The causal variants are themselves the disease determinant; there are no known separate susceptibility loci. Documented disease alleles include (non-exhaustive): - *ITGB3* ($\beta 3$): **p.Asp723His (D723H)**, **p.Thr720del (T720del)**, D749H, T746P, H748P, R760C, plus membrane-proximal missense/deletion variants (e.g., C560R, $\beta\text{TD_del p.647-686}$ in Glanzmann-like macrothrombocytopenia). - *ITGA2B* (αIIb): **p.Arg995Trp (R995W)**, R995Q, **R1026W**, R1026Q, G1007V.

Environmental / demographic risk factors. As a monogenic dominant disorder, the disease is **not caused by environmental exposures**. Relevant non-genetic modifiers of *bleeding severity* (not disease causation) are standard hemostatic challenges: surgery, trauma, dental extraction, childbirth/postpartum, menstruation, and antiplatelet/anticoagulant drug use. Female sex confers additional gynecologic/obstetric bleeding burden (menorrhagia, postpartum hemorrhage). A **positive family history** is a hallmark given dominant transmission.

Protective factors. No specific genetic or environmental protective factors are established. General avoidance of antiplatelet agents (aspirin, NSAIDs) reduces bleeding risk. There is no evidence of a protective allele.

Gene–environment interactions. No formal GxE interaction has been characterized. Practically, the penetrant genetic lesion sets a fixed platelet phenotype whose clinical expression is *unmasked* by hemostatic stressors (surgery, trauma, menstruation, delivery).

3. Phenotypes

BDPLT16 phenotypes comprise **laboratory abnormalities** and **clinical bleeding signs/symptoms**. Bleeding is typically **mild-to-moderate**, lifelong, and mucocutaneous; severity is **variable** even within families.

Phenotype	Type	Characteristics	Suggested HPO term
Thrombocytopenia	Lab abnormality	Congenital, lifelong, stable; reduced platelet count	HP:0001873 Thrombocytopenia
Large/giant platelets (macrothrombocytopenia)	Lab/morphologic	Enlarged round platelets on smear; increased anisotropy	HP:0040326 Increased mean platelet volume
Reduced α IIb β 3 surface expression	Lab abnormality	Low (not absent) GPIIb/IIIa by flow cytometry	HP:0011876 Abnormal platelet function
Impaired platelet aggregation	Lab abnormality	Reduced (not absent) response to ADP, collagen, TRAP-6; ristocetin normal	HP:0003540 Impaired platelet aggregation
Reduced ATP/dense-granule release	Lab abnormality	Low ATP release to physiologic agonists	HP:0011876 Abnormal platelet function
Enlarged/fused α -granules	Ultrastructural	Abnormal large α -granules, some giant/fused	—
Easy bruising / mucocutaneous bleeding	Clinical sign	Mild-moderate, episodic (provoked by trauma/surgery)	HP:0000978 Bruising susceptibility
Epistaxis	Clinical sign	Recurrent, mild-moderate	HP:0000421 Epistaxis
Gingival/gum bleeding	Clinical sign	Mucosal	HP:0000225 Gingival bleeding
Menorrhagia / heavy menstrual bleeding	Clinical sign	Prominent in affected women; may be chronic	HP:0000132 Abnormal menstruation (menorrhagia)
Prolonged bleeding after surgery/trauma	Clinical sign	Episodic, provoked	HP:0004846 Abnormal bleeding

Age of onset. Congenital laboratory phenotype (thrombocytopenia/large platelets present from birth); bleeding symptoms emerge in **childhood** and persist lifelong. **Severity** is mild-to-moderate and **variable**; **progression** is stable (non-progressive), with bleeding **episodic** and provoked by hemostatic challenge.

Frequency among affected individuals. In the largest series (10 Portuguese GTLS families, 33 patients), the core lab tetrad — macrothrombocytopenia, low α Ib β 3 expression, impaired aggregation/ATP release, and low PAC-1 binding — was essentially universal, while clinical bleeding ranged from **absent to moderate** (PMID: 33276370).

Quality of life impact. Generally modest given mild-moderate severity, but recurrent epistaxis and especially **menorrhagia** can meaningfully impair quality of life and cause iron-deficiency anemia; peri-operative and peri-partum periods carry the highest morbidity risk. No disease-specific EQ-5D/SF-36 data are available.

4. Genetic / Molecular Information

Causal genes. *ITGA2B* (α Ib; HGNC:6138; OMIM 607759; 17q21.31) and *ITGB3* (β 3; HGNC:6156; OMIM 173470; 17q21.32). The two chains form the heterodimeric platelet fibrinogen receptor α Ib β 3 (GPIIb/IIIa, integrin α Ib β 3), expressed at ~60,000–80,000 copies per platelet in mouse and abundantly in human platelets (PMID: 11154120).

Pathogenic variants (representative).

Gene	Variant	Type	Effect
<i>ITGB3</i>	p.Asp723His (D723H)	Missense	Breaks α IIb-R995/ β 3-D723 salt bridge → constitutive activation
<i>ITGB3</i>	p.Thr720del (T720del)	In-frame deletion	Spontaneous α IIb β 3 activation
<i>ITGB3</i>	D749H, T746P, H748P, R760C	Missense	Membrane-proximal, activating
<i>ITGB3</i>	C560R (EGF-like domain)	Missense	Constitutive activation, differing surface expression
<i>ITGB3</i>	β TD_del (p.647-686)	In-frame deletion	Constitutive activation (β -tail domain)
<i>ITGA2B</i>	p.Arg995Trp/Gln (R995W/Q)	Missense	Breaks salt bridge → constitutive activation
<i>ITGA2B</i>	R1026W, R1026Q, G1007V	Missense	Membrane-proximal, activating

Variant classification. ClinVar corroborates BDPLT16 alleles: *ITGB3* Asp723His and *ITGA2B* Arg995 (R995W/R995Q) records are present, and ~8 *ITGA2B* records are annotated to "macrothrombocytopenia" (Finding F008). The majority of *ITGA2B* ($\approx$ 345) and *ITGB3* ($\approx$ 266) pathogenic ClinVar entries reflect the **allelic recessive Glanzmann thrombasthenia**, not the rare dominant BDPLT16.

Variant type/class. Predominantly **missense**; also **small in-frame deletions** (T720del, β TD_del). All cluster in the membrane-proximal transmembrane/cytoplasmic interface.

Allele frequency. Disease alleles are **private/ultra-rare** and effectively **absent from gnomAD**, consistent with a highly penetrant dominant deleterious effect (Finding F008).

Somatic vs germline. **Germline**, heterozygous, dominantly inherited.

Functional consequence. **Gain-of-function** (constitutive integrin activation) — mechanistically opposite to the loss-of-function of classic GT. In-silico modeling shows mutated residues "directly modify the salt bridge linking the intra-cytoplasmic part of α IIb to β 3" ([PMID: 29090484](#)).

Modifier genes / epigenetics / chromosomal abnormalities. No specific modifier genes, epigenetic marks, or large-scale chromosomal abnormalities are established for BDPLT16. The lesion is a point mutation / small in-frame deletion, not a copy-number or structural variant.

5. Environmental Information

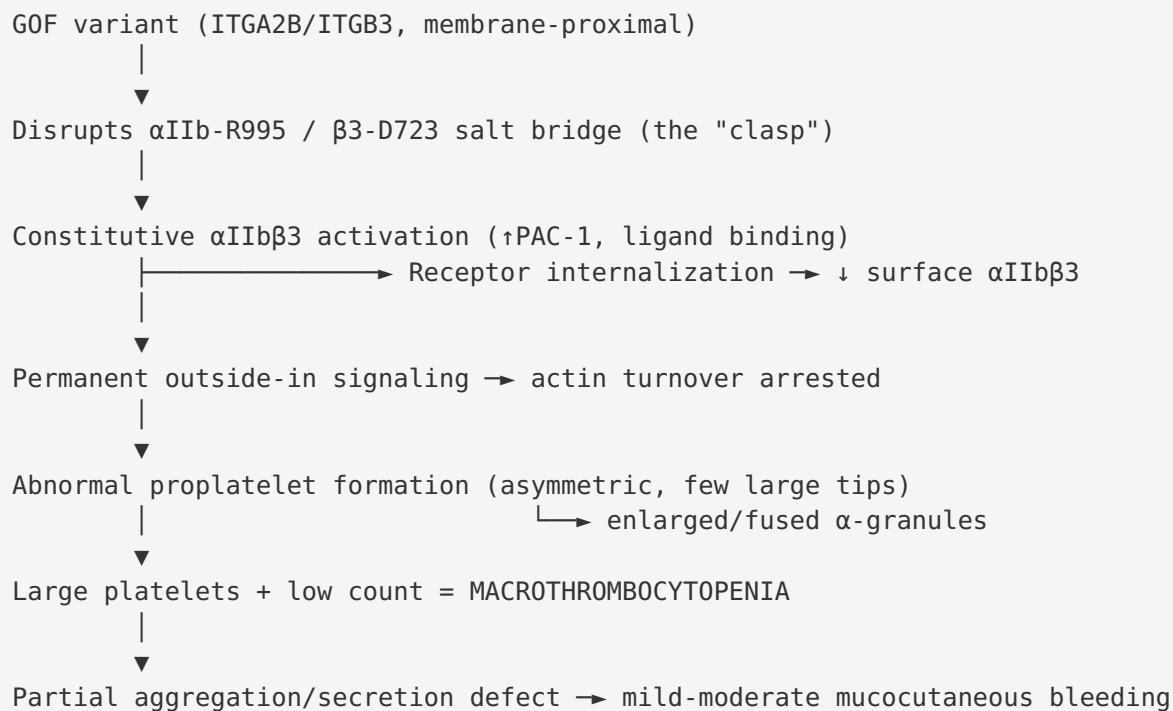
BDPLT16 is a **monogenic disorder with no environmental, lifestyle, or infectious etiology**. No toxins, radiation, occupational exposures, dietary factors, or pathogens cause or trigger the disease. Environmental factors are relevant only as **bleeding precipitants** (trauma, surgery, dental procedures, childbirth) and as **aggravators** via antiplatelet/anticoagulant drug exposure (aspirin, NSAIDs). Not applicable: infectious agents.

6. Mechanism / Pathophysiology

Ordered causal chain

1. A heterozygous **gain-of-function variant** arises in *ITGA2B* (α IIb) or *ITGB3* (β 3) in the membrane-proximal region → **disrupts the conserved α IIb-Arg995 ↔ β 3-Asp723 intracytoplasmic salt bridge** (the "clasp" that holds the integrin in its resting/bent state). (*Demonstrated by in-silico + functional studies.*)
2. Loss of the salt-bridge clasp **releases the cytoplasmic/transmembrane restraint** → the integrin adopts an **extended, constitutively active conformation** → **spontaneous (agonist-independent) α IIb β 3 activation** with increased binding of activation-specific ligands (↑ PAC-1 binding; adhesion to fibrinogen/VWF under shear). (*Demonstrated in CHO/293T transfectants and patient platelets.*)
3. Constitutive ligand binding triggers **permanent "outside-in" signaling** → **arrest of actin turnover at the polymerization stage** and disordered cytoskeletal (actin/tubulin) reorganization in megakaryocytes. (*Demonstrated.*)
4. Cytoskeletal perturbation **impairs proplatelet formation** → megakaryocytes generate **abnormal cytoplasmic extensions / asymmetric "barbell" proplatelets with fewer, larger tips**. (*Demonstrated in patient CD34⁺ MK cultures and cell models.*)

5. Abnormal proplatelet fragmentation **leads to incorrect platelet sizing and reduced platelet output** → **macrothrombocytopenia** (large platelets, low count). (*Demonstrated.*)
6. **Branch — receptor internalization:** constitutive activation promotes **α IIb β 3 internalization** → **reduced surface expression** of the receptor on circulating platelets. (*Demonstrated.*)
7. **Branch — granule abnormality:** disordered biogenesis yields **enlarged/fused α -granules** (some giant). (*Observed.*)
8. Reduced receptor density + dysfunctional granule release + abnormal cytoskeleton → **partial platelet aggregation and secretion defect** (reduced but not absent responses to ADP, collagen, TRAP-6) → **mild-to-moderate mucocutaneous bleeding** when hemostasis is challenged. (*Demonstrated clinically.*)



Detail by category

- **Molecular pathways / biochemical abnormality.** The defect is **integrin bidirectional (inside-out/outside-in) signaling**. The proximate biochemical lesion is loss of the electrostatic salt bridge that maintains the low-affinity resting state; the downstream signaling engages Rap1/talin-associated outside-in activation cascades and actin dynamics ([PMID: 29380037](#); [PMID: 26452979](#)).

- **Protein dysfunction. Gain-of-function** conformational change: the mutant integrin is trapped in an active/extended state rather than misfolded or degraded. This distinguishes it from GT loss-of-function (absent/nonfunctional receptor).
- **Cellular processes.** Dysregulated **cytoskeletal reorganization** (actin polymerization/turnover, tubulin-dependent protrusions), **abnormal proplatelet formation**, and **receptor endocytosis/internalization**.
- **Cell types involved.** **Megakaryocytes** (CL:0000556) as the site of defective platelet production; **platelets/thrombocytes** (CL:0000233) as the dysfunctional end product.
- **Suggested GO terms.** GO:0007229 integrin-mediated signaling pathway; GO:0033628 regulation of cell adhesion mediated by integrin; GO:0030168 platelet activation; GO:0030220 platelet formation; GO:0007010 cytoskeleton organization; GO:0006897 endocytosis. Cellular component: GO:0009986 cell surface, GO:0031093 platelet alpha granule, GO:0015629 actin cytoskeleton.
- **Immune / metabolic / infectious involvement.** None intrinsic. (Misdiagnosis as *immune* thrombocytopenia is a clinical, not mechanistic, issue.)

Instructive natural experiment (sitosterolemia). In a murine sitosterolemia model, plant-sterol accumulation in the platelet membrane produced **constitutive fibrinogen binding to α IIb β 3, receptor internalization, and macrothrombocytopenia** — phenocopying, via a non-genetic route, the same "constitutively active α IIb β 3 $\rightarrow$ macrothrombocytopenia" logic and reinforcing causality ([PMID: 23926302](#)).

7. Anatomical Structures Affected

- **Organ/system level.** Primary involvement is the **hematopoietic system / blood** (UBERON:0000178 blood) and **bone marrow** (UBERON:0002371) where megakaryopoiesis occurs. The **mucocutaneous vasculature** (skin, nasal mucosa, gingiva, gastrointestinal and genitourinary mucosa) is the site of bleeding manifestations. Secondary involvement: **endometrium/uterus** (menorrhagia) and potential iron-deficiency anemia affecting systemic function.
- **Tissue/cell level.** **Megakaryocytes** (CL:0000556) in bone marrow; **platelets/thrombocytes** (CL:0000233) in circulation. No solid-organ parenchymal tissue is primarily targeted.

- **Subcellular level. Plasma membrane / cell surface** (integrin localization; GO:0005886), **platelet α -granules** (GO:0031093; enlarged/fused), and the **actin cytoskeleton** (GO:0015629).
 - **Localization / lateralization.** Systemic (blood-borne); bleeding sites are **not lateralized** — diffuse mucocutaneous. No focal or asymmetric anatomic lesion.
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8. Temporal Development

- **Onset. Congenital** laboratory phenotype (thrombocytopenia and large platelets from birth); clinical bleeding typically recognized in **childhood**. Onset pattern is **chronic/insidious** rather than acute.
 - **Progression. Stable / non-progressive**; the platelet count and morphology remain relatively constant over life. Bleeding is **episodic**, provoked by hemostatic challenges (surgery, trauma, menstruation, delivery).
 - **Disease course / duration. Chronic, lifelong.** There is no staging system and no natural tendency to remission or worsening.
 - **Critical periods.** Windows of heightened vulnerability are **surgical/dental procedures, trauma, menarche/menstruation, pregnancy and the peripartum period** — the key opportunities for prophylactic hemostatic intervention.
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9. Inheritance and Population

- **Inheritance pattern. Autosomal dominant** (heterozygous gain-of-function). Contrast with recessive classic GT.
- **Epidemiology.** No formal prevalence/incidence figures exist; BDPLT16 is **ultra-rare**, reported as a handful of families worldwide. Known pedigrees include the original *ITGB3* D723H family (5 affected over 3 generations; [PMID: 18065693](#)), Japanese T720del and related families ([PMID: 29380037](#)), French salt-bridge families ([PMID: 29090484](#)), and a Portuguese cohort of 10 families/33 patients ([PMID: 33276370](#)).
- **Penetrance / expressivity.** Cosegregation is strong (the laboratory phenotype tracks with the variant across generations), implying **high penetrance for the lab phenotype**;

bleeding severity is variably expressed (absent-to-moderate) even among carriers of the same allele.

- **Anticipation / mosaicism / consanguinity / founder effects.** No genetic anticipation (not a repeat-expansion disorder). No consanguinity required (dominant). The Portuguese cluster of 10 families may reflect ascertainment and/or shared alleles, but no formal founder haplotype has been established. Germline mosaicism not specifically documented.
- **Carrier frequency.** Not applicable in the recessive sense; disease alleles are private/ultra-rare and essentially absent from gnomAD (Finding F008).
- **Population demographics.** No strong ethnic predilection; cases reported across European (French, Portuguese, Italian) and Asian (Japanese) populations. **Sex ratio** for the genetic trait is ~1:1 (autosomal); affected **women carry additional gynecologic/obstetric bleeding burden**. Age distribution spans all ages given lifelong congenital nature.

10. Diagnostics

Laboratory tests. - **CBC with peripheral blood smear:** thrombocytopenia with **large/giant platelets**; increased mean platelet volume and platelet anisotropy. Automated counters may **undercount** large platelets. - **Flow cytometry:** **reduced (not absent) α IIB β 3/GPIIb-IIIa surface expression; reduced PAC-1 binding** upon stimulation (TRAP-6, ADP) — reflecting low activation-inducible binding sites. - **Light transmission aggregometry (LTA): reduced but not absent** aggregation to ADP and collagen; **normal ristocetin-induced agglutination** (distinguishing from Bernard-Soulier/VWD); reduced ATP/dense-granule release. - **Electron microscopy (specialized):** enlarged, sometimes fused α -granules; abnormal platelet morphology.

Biomarkers. The defining "biomarker" is the **combination** of macrothrombocytopenia + reduced (not absent) α IIB β 3 + partial aggregation defect + activating membrane-proximal *ITGA2B/ITGB3* variant.

Genetic testing (definitive). - **Recommended approach:** NGS-based **inherited platelet disorder gene panels** covering *ITGA2B*, *ITGB3*, and the broader macrothrombocytopenia genes; **WES/WGS** are useful when panels are non-diagnostic. Confirm candidate variants by **Sanger sequencing** and test **family segregation**. - **Single-gene testing** of *ITGA2B/ITGB3* is appropriate when phenotype is strongly suggestive. - CMA/karyotype/FISH/mtDNA/repeat-expansion testing are **not applicable** (point-mutation disorder).

Imaging / electrophysiology / biopsy. Not diagnostic; imaging is used only to evaluate bleeding complications. Bone marrow examination is typically normal (normal megakaryocyte numbers), useful mainly to exclude other causes.

Clinical criteria & differential diagnosis. No formal consensus criteria; diagnosis is integrative (phenotype + genetics). **Differential diagnosis** of dominant macrothrombocytopenia with platelet dysfunction includes:

Condition	Inheritance	Key distinguishing feature
BDPLT16 (this disease)	AD (GOF)	Reduced (not absent) α IIb β 3; partial aggregation defect; activating membrane-proximal <i>ITGA2B/ITGB3</i> variant
Classic Glanzmann thrombasthenia	AR (LOF)	Absent α IIb β 3/aggregation; normal count & size
Bernard-Soulier syndrome	AR	GPIb-IX-V defect; abnormal ristocetin agglutination
MYH9-related disorders (May-Hegglin)	AD	Leukocyte Döhle-like inclusions; <i>MYH9</i> variant
von Willebrand disease	AD/AR	VWF defect; abnormal ristocetin; corrects with VWF
Immune thrombocytopenia (ITP)	Acquired	No family history; normal platelet size; steroid-responsive

Congenital macrothrombocytopenias "share common clinical and laboratory features... and patients are often misdiagnosed with and treated for idiopathic thrombocytopenic purpura" (PMID: 16169642).

Screening. No population/newborn screening. **Cascade genetic testing** of at-risk relatives is appropriate once a familial variant is identified; prenatal/preimplantation testing is technically feasible but rarely indicated given the mild phenotype.

11. Outcome / Prognosis

- **Survival / life expectancy.** **Normal life expectancy;** BDPLT16 is not associated with increased mortality in the absence of catastrophic bleeding. No disease-specific mortality data.

- **Morbidity.** Driven by bleeding: recurrent epistaxis, easy bruising, **menorrhagia** (with risk of **iron-deficiency anemia**), and peri-operative/peri-partum hemorrhage. Serious spontaneous bleeding (e.g., intracranial) is rare in mild-moderate disease, though documented in the severe allelic condition GT.
 - **Disease course.** Chronic and stable; complications are episodic and largely preventable with appropriate hemostatic management around challenges.
 - **Prognostic factors.** Bleeding phenotype severity, sex (gynecologic burden), exposure to antiplatelet drugs, and the nature of hemostatic challenges. No validated molecular prognostic biomarker; there is a broad genotype–phenotype trend (all activating salt-bridge variants → macrothrombocytopenia), but bleeding severity is variably expressed.
 - **Quality-of-life measures.** No disease-specific EQ-5D/SF-36/PROMIS data; QoL is generally good but reduced in those with significant menorrhagia.
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12. Treatment

Overarching principle. There is **no BDPLT16-specific or curative therapy**. Management is **supportive/on-demand**, extrapolated from inherited platelet function disorders and the allelic condition Glanzmann thrombasthenia.

Pharmacotherapy and hemostatic agents.

Intervention	Role	NCIT suggestion
Tranexamic acid / antifibrinolytics	First-line for mucosal bleeding, menorrhagia, dental/minor surgery	NCIT:C739 Tranexamic Acid
Recombinant activated factor VII (rFVIIa)	Major bleeds/surgery; also reduces risk of platelet alloimmunization	NCIT:C1836 Recombinant Factor VIIa
Platelet transfusion	Severe/life-threatening bleeding or major surgery; use judiciously (alloimmunization risk)	NCIT:C15328 Platelet Transfusion
Hormonal therapy (combined OCP, progestins, LNG-IUS)	Control of heavy menstrual bleeding	NCIT:C548 Hormone Therapy
Desmopressin (DDAVP)	Considered in mild platelet disorders; variable benefit	NCIT:C29179 Desmopressin
Iron supplementation	Treat/prevent iron-deficiency anemia from chronic blood loss	NCIT:C1381 Iron Supplement
Local measures	Pressure, nasal packing, topical agents for epistaxis	—

Evidence: in GT, "bleeding control was achieved through the use of antifibrinolytic agents and recombinant factor VIIa, which also reduces the risk of platelet alloimmunization" ([PMID: 41778036](#)). In women, "management of chronic HMB required a combination therapy including antifibrinolytics (tranexamic acid [TXA]), hormonal therapies, and recombinant factor VIIa (rFVIIa)" ([PMID: 40123272](#)). Local measures, desmopressin, and antifibrinolytics are first-line for mild inherited platelet disorders ([PMID: 23269640](#)).

Advanced / experimental therapeutics. No approved gene therapy, cell therapy, RNA-based, targeted, or immunotherapy for BDPLT16. Caution with anticoagulation — provoked thrombosis can occur in $\alpha\text{IIb}\beta\text{3}$ disorders and carries a narrow therapeutic window ([PMID: 41591555](#), reported in GT).

Pharmacogenomics. Avoid antiplatelet drugs (aspirin, NSAIDs, P2Y₁₂ inhibitors). No BDPLT16-specific pharmacogenomic guidance.

Treatment strategy. Individualized, challenge-based prophylaxis: pre-procedural antifibrinolytics ± rFVIIa ± platelets; peri-partum planning in a hemophilia/bleeding-disorder center; proactive management of menorrhagia and iron status.

13. Prevention

- **Primary prevention.** Not possible (genetic). Preventing *bleeding events*: avoid antiplatelet/anticoagulant drugs, use careful surgical/dental planning with prophylactic hemostatic cover.
 - **Secondary prevention.** Early diagnosis (correctly distinguishing from ITP to avoid unnecessary steroids/splenectomy), **cascade genetic testing** of relatives, and iron-deficiency screening in menorrhagic patients.
 - **Tertiary prevention.** Prevent complications: peri-operative/peri-partum hemostatic protocols, minimize platelet transfusions to reduce **alloimmunization** (favor rFVIIa where possible), treat anemia.
 - **Genetic counseling.** Autosomal dominant → **50% transmission risk** to offspring; counsel on variable bleeding expressivity; offer family-cascade testing; prenatal/PGT feasible but seldom pursued given mild phenotype.
 - **Immunization / public health / environmental interventions.** Not applicable beyond general bleeding-risk counseling.
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14. Other Species / Natural Disease

- **Taxonomy / orthologs.** Mouse orthologs *Itga2b* (NCBI Gene 16399) and *Itgb3* (NCBI Gene 16416). Murine $\alpha\text{IIb}\beta 3$ is functionally homologous to human, expressed at ~60,000–80,000 copies/platelet, with conserved EDTA-dissociation and activation biology ([PMID: 11154120](#)).
- **Natural disease in other species.** No well-characterized **naturally occurring, dominant gain-of-function** $\alpha\text{IIb}\beta 3$ macrothrombocytopenia has been catalogued in companion animals for BDPLT16 specifically; naturally occurring *Glanzmann thrombasthenia* (loss-of-function) is documented in dogs and horses (OMIA), which is the allelic recessive disease rather than BDPLT16.

- **Comparative biology.** The α IIb-R995/ β 3-D723 salt bridge and integrin activation machinery are evolutionarily conserved, so the mechanistic principle (salt-bridge disruption → constitutive activation) is expected to be conserved across mammals.
- **Zoonotic potential.** None (non-transmissible genetic disorder).

15. Model Organisms

In vitro / cellular models (primary evidence base for BDPLT16). - Transfected cell lines (CHO, HEK293/293T) expressing mutant α IIb β 3 (β 3-D723H/H723, β 3- β TD_del p.647-686, β 3-C560R, β 3-T720del) reproduce **constitutive activation, increased PAC-1 binding, adhesion to fibrinogen/VWF under shear**, and formation of **abnormal proplatelet-like cytoplasmic extensions** not seen with wild-type (PMID: 18065693; PMID: 25806962; PMID: 29380037; PMID: 26452979). - **Patient-derived CD34+ stem-cell megakaryocyte cultures** confirm **abnormal proplatelet formation** in the propositus, directly linking the variant to the production defect (PMID: 18065693).

Mouse models. - The **β 3-integrin knockout (*Itgb3*^{-/-})** mouse models **loss-of-function Glanzmann thrombasthenia**, NOT gain-of-function BDPLT16: "The mice are viable and fertile, and show all the cardinal features of GT (defects in platelet aggregation and clot retraction, prolonged bleeding times, and cutaneous and gastrointestinal bleeding)," plus placental defects and reduced survival (PMID: 9916135). - **Sitosterolemia mouse (*Abcg5/Abcg8*^{-/-})** provides a **phenocopy of the mechanism** — sterol-induced constitutive α IIb β 3 activation, internalization, and macrothrombocytopenia — even though the genetic cause differs (PMID: 23926302).

Phenotype recapitulation & limitations. No published knock-in mouse carrying a human BDPLT16 activating salt-bridge variant is established; thus the dominant gain-of-function macrothrombocytopenia is **best modeled in vitro and in patient megakaryocytes**, while mouse knockouts capture only the allelic loss-of-function disease. A conditional/knock-in *Itgb3* D723H (or *Itga2b* R995W) mouse is an obvious gap.

Mechanistic Model / Interpretation

BDPLT16 is best understood as a **"stuck-on" integrin disorder**. The $\alpha\text{IIb}\beta 3$ fibrinogen receptor normally rests in a bent, low-affinity conformation held by an intracellular clasp — the **$\alpha\text{IIb-R995} \leftrightarrow \beta 3\text{-D723}$ salt bridge**. BDPLT16 variants **break this clasp**, so the receptor is **constitutively active** even without agonist. Paradoxically, this gain-of-function produces **bleeding**, because (a) chronic activation triggers **receptor internalization**, lowering surface density; and (b) permanent **outside-in signaling** freezes the megakaryocyte cytoskeleton, **garbling proplatelet formation** so platelets emerge **too large and too few** with impaired secretion. The result is a **macrothrombocytopenia with a partial functional defect** — the mirror image of Glanzmann thrombasthenia, where the same receptor is simply **absent/nonfunctional** and platelet number/size are normal.

This unifying logic explains every observed feature: dominant inheritance (a single active allele poisons proplatelet formation), reduced-not-absent $\alpha\text{IIb}\beta 3$ (internalization vs. deletion), partial-not-absent aggregation, large fused α -granules, and mild-to-moderate bleeding. The sitosterolemia phenocopy independently validates the causal step "constitutive $\alpha\text{IIb}\beta 3$ activation $\rightarrow$ internalization $\rightarrow$ macrothrombocytopenia."

BDPLT16 vs. classic Glanzmann thrombasthenia (allelic contrast).

Feature	BDPLT16	Classic Glanzmann thrombasthenia
Molecular effect	Gain-of-function (constitutive activation)	Loss-of-function (absent/defective receptor)
Inheritance	Autosomal dominant	Autosomal recessive
$\alpha\text{IIb}\beta 3$ surface level	Reduced (not absent)	Absent/markedly reduced
Platelet count	Low (thrombocytopenia)	Normal
Platelet size	Large (macrothrombocytes)	Normal
Aggregation defect	Partial	Absent/severe
Variant location	Membrane-proximal TM/cytoplasmic	Throughout gene
Bleeding severity	Mild-moderate	Moderate-severe

Evidence Base

PMID	Contribution	Supports
29090484	Salt-bridge disruption (R995W, D723H); enlarged α -granules; mild-moderate phenotype	F001, F003
29380037	$\beta 3$ T720del; membrane-proximal clustering; constitutive activation	F001, F003
33276370	10 families/33 patients; 7 variants; defines core clinical/lab phenotype	F001, F003
26452979	Cytoskeletal mechanism: outside-in signaling arrests actin turnover; reduced surface expression	F001, F002
18065693	Original D723H pedigree; PAC-1 $\uparrow$; CHO model; patient MK abnormal proplatelets	F002, F004
25806962	β TD_del and C560R cause abnormal cytoplasmic extensions (proplatelet defect)	F002, F004
23926302	Sitosterolemia: sterol-induced constitutive α IIb β 3 activation $\rightarrow$ macrothrombocytopenia (mechanism phenocopy)	F002
9916135	$\beta 3$ -null mouse = loss-of-function GT model (not BDPLT16)	F004
11154120	Murine α IIb β 3 structure/function homology	F004
16169642	Congenital macrothrombocytopenias misdiagnosed as ITP	F005
34400424	Differential diagnosis framework (BSS, MYH9, GT, VWD)	F005
41778036	Antifibrinolytics + rFVIIa hemostatic management	F006
40123272	Menorrhagia management (TXA, hormonal, rFVIIa)	F006
23269640	First-line local measures/DDAVP/antifibrinolytics in mild platelet disorders	F006
41503871	Review defining ITGA2B/ITGB3-related macrothrombocytopenia as GOF subset	F001

Verbatim supporting quotes. - "In silico analysis shows how the two mutated amino acids directly modify the salt bridge linking the intra-cytoplasmic part of α Ib to β 3 of the integrin α Ib β 3." — PMID: 29090484 - "Reported mutations were highly clustered at the membrane proximal region of α Ib β 3, which affected the critical interaction between α Ib R995 and β 3 D723, resulting in a constitutionally active form of the α Ib β 3 complex." — PMID: 29380037 - "the constitutive activation of the α Ib β 3-H723 receptor causes abnormal proplatelet formation, leading to incorrect sizing of platelets and the thrombocytopenia observed in the pedigree." — PMID: 18065693 - "permanent triggering of α Ib β 3-mediated outside-in signaling causes an impairment of cytoskeletal reorganization arresting actin turnover at the stage of polymerization." — PMID: 26452979 - "Patients had absent to moderate bleeding, macrothrombocytopenia, low α Ib β 3 expression, impaired platelet aggregation/ATP release to physiological agonists and low expression of activation-induced binding sites on α Ib β 3 (PAC-1)." — PMID: 33276370 - "Many of these disorders share common clinical and laboratory features, making accurate diagnosis difficult and patients are often misdiagnosed with and treated for idiopathic thrombocytopenic purpura." — PMID: 16169642 - "Bleeding control was achieved through the use of antifibrinolytic agents and recombinant factor VIIa, which also reduces the risk of platelet alloimmunization." — PMID: 41778036

Limitations and Knowledge Gaps

1. **No formal epidemiology.** Prevalence/incidence are unknown; evidence is a handful of families. Penetrance and expressivity estimates are qualitative.
2. **No in vivo gain-of-function model.** No knock-in mouse (e.g., *Itgb3* D723H or *Itga2b* R995W) has been established; mechanism relies on in vitro transfectants and patient megakaryocytes. Mouse *Itgb3*^{-/-} models only the allelic loss-of-function GT.
3. **Mechanistic exceptions.** Recent reports of **non-activating *ITGB3* variants** causing macrothrombocytopenia challenge the uniform gain-of-function model (PMID: 41503871), indicating additional/alternative mechanisms remain to be defined.
4. **Treatment evidence is extrapolated**, not BDPLT16-specific — drawn from GT and general inherited platelet disorders; no trials in BDPLT16.
5. **No QoL, natural-history, or prognostic-biomarker data** specific to BDPLT16.
6. **Genotype–phenotype correlation** for bleeding severity is weak; modifier genes are unidentified.

Proposed Follow-up Experiments / Actions

1. **Generate a knock-in mouse** (*Itgb3* p.D723H or *Itga2b* p.R995W) or iPSC-derived megakaryocyte lines to model dominant gain-of-function BDPLT16 in vivo and quantify proplatelet defects.
 2. **Systematic genotype–phenotype registry** across reported families to estimate penetrance, expressivity, and bleeding-severity predictors.
 3. **Mechanistic dissection of non-activating *ITGB3* macrothrombocytopenia variants** to determine whether a distinct (activation-independent) pathway contributes.
 4. **Single-cell transcriptomics/proteomics of patient megakaryocytes** to map the cytoskeletal and granule-biogenesis programs perturbed by constitutive $\alpha\text{IIb}\beta 3$ signaling (GO:0030220 platelet formation).
 5. **Prospective evaluation of hemostatic strategies** (antifibrinolytics $\pm$ rFVIIa vs. platelet transfusion) to build BDPLT16-specific, alloimmunization-sparing protocols.
 6. **ClinVar/ACMG reclassification effort** to curate BDPLT16 alleles distinctly from recessive GT alleles, improving diagnostic reporting.
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Report compiled from 8 confirmed findings and 28 reviewed papers across 5 investigation iterations. Evidence types: human clinical pedigrees/case series, in vitro heterologous-cell and patient-megakaryocyte studies, and mouse models. Ontology suggestions (HPO, GO, CL, UBERON, NCIT) are provided throughout for knowledge-base population.
